

U-Net vs. SAM: Semantic Segmentation on PASCAL VOC 2012

This project trains and evaluates a U-Net model for semantic segmentation on the PASCAL VOC 2012 dataset. The goal was to compare the trained U-Net's performance against the zero-shot inference capabilities of the Segment Anything Model (SAM).

Dataset

- Dataset:** PASCAL VOC 2012
- Training Set:** 1,464 images
- Validation Set:** 1,449 images

Part 1: U-Net Model Training & Results

A U-Net model was trained from scratch for 50 epochs to perform segmentation on the VOC dataset.

- Architecture:** Standard U-Net
- Loss Function:** A custom `CombinedLoss` (50% Cross-Entropy + 50% Dice Loss) was used to handle class imbalance and improve segmentation boundaries.
- Optimizer:** Adam
- Learning Rate:** 1e-4, with a `ReduceLROnPlateau` scheduler.

U-Net Performance

The U-Net model trained successfully and achieved a stable performance on the validation set.

Final Validation Metrics:

Metric	Value
Validation Loss	0.582
Validation IoU	0.629
Validation Accuracy	0.908

Training Plots

The following graphs show the model's convergence over 50 epochs.

IoU (Intersection over Union) and Loss:

Accuracy:

Part 2: SAM (Segment Anything Model) Analysis - ABORTED

The second phase of this project was to run the pre-trained SAM (`vit_b` checkpoint) on the same 1,449-image validation set to benchmark its zero-shot performance.

Status: This analysis was **not completed**.

Reason for Failure

An issue was identified in the data-loading step for the SAM inference script. The script was incorrectly configured to load **all 17,112 images** from the root `JPEGImages` directory, rather than being filtered to the 1,449 images specified in the `val.txt` file.

This error made the inference process computationally unfeasible. The estimated processing time for the entire dataset was projected to be over 700 hours. Due to these extreme time and computational constraints, the inference run was aborted.

Conclusion

The U-Net model provides a strong baseline for segmentation on the PASCAL VOC dataset, achieving over **62% Mean IoU**. A direct performance comparison with SAM was not possible due to the data-loading configuration error and resulting prohibitive runtime.